

i2DGS: Inverse 2D Gaussian Splatting for OLAT

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Abstract

We propose i2DGS, a novel method for high-fidelity 3D geometry and material reconstruction from multi-view One-Light-at-a-Time (OLAT) images. To our knowledge, i2DGS is the first approach to combine albedo, roughness, and metallic map estimation with 2D Gaussian Splatting (2DGS) through differentiable physically-based rendering (PBR). Our two-stage inverse rendering pipeline first reconstructs geometry via standard 2DGS on uniformly-lit images, then refines material properties under OLAT supervision using adapted PBR primitives. We further introduce a fully differentiable BVH-accelerated ray tracing module for shadow modeling. Experiments on synthetic and real-world OLAT captures demonstrate that i2DGS outperforms comparable methods in both geometry and material reconstruction, producing high-fidelity PBR assets suitable for relighting.

1. Introduction

The recovery of physically-based rendering (PBR) materials from multi-view images remains a fundamental challenge in computer vision and graphics. Representations such as Neural Radiance Fields (NeRF) [14] and 3D Gaussian Splatting (3DGS) [11] have advanced novel view synthesis, yet they intrinsically entangle lighting, materials, and geometry, and often fail to reconstruct precise underlying surfaces. 2D Gaussian Splatting (2DGS) [9] mitigates these geometric limitations through an explicit surfel-based representation, but its geometry is optimized for view synthesis rather than physical correctness, making high-quality mesh extraction difficult. For downstream use in game engines and film pipelines, both accurate surface reconstruction and the explicit recovery of spatially varying BRDF (SVBRDF) properties—albedo, roughness, and metalness—are essential.

A successful line of research addresses these geometric ambiguities by integrating Multi-View Photometric Stereo (MVPS) principles into neural fields, as in PS-NeRF [19]. These methods leverage images captured under varying illumination to jointly solve for geometry and materials, but

inherit NeRF’s prohibitively slow training and rendering. More recently, PS-GS [5] brought photometric stereo to the 2DGS framework. While PS-GS achieves strong results, it treats photometric stereo as an external geometric regularizer—fitting a neural lighting model to normals derived from uncalibrated images—and thus only loosely couples geometry and material estimation rather than modeling their physical codependence.

We propose i2DGS, a novel inverse rendering pipeline that directly couples geometry and material estimation through differentiable physically-based rendering. Our two-stage approach first establishes a robust geometric initialization using 2DGS on uniformly lit images, then jointly optimizes geometry and PBR material parameters under One-Light-At-a-Time (OLAT) supervision with adapted PBR primitives.

Our main contributions are as follows:

- A two-stage inverse rendering pipeline that combines the surface reconstruction quality of 2D Gaussian Splatting with joint PBR material optimization under OLAT supervision.
- A fully differentiable, BVH-accelerated ray tracing module for shadow modeling that dynamically adapts to scene density.
- Experiments on synthetic OLAT captures showing that i2DGS produces high-fidelity relightable assets and outperforms comparable methods in both geometry and material reconstruction.

2. Related Work

2.1. Novel View Synthesis

Neural Radiance Fields (NeRF) [14] marked a breakthrough in novel view synthesis by parameterizing scenes as continuous 5D functions. Subsequent works addressed aliasing (Mip-NeRF [1]) and training speed (Instant NGP [15]). 3D Gaussian Splatting (3DGS) [11] offered an explicit alternative, achieving real-time rendering through tile-based rasterization of 3D Gaussians. Extensions of 3DGS target temporal dynamics [13] and geometric accuracy, notably SuGaR [7] and 2D Gaussian Splatting (2DGS) [9].

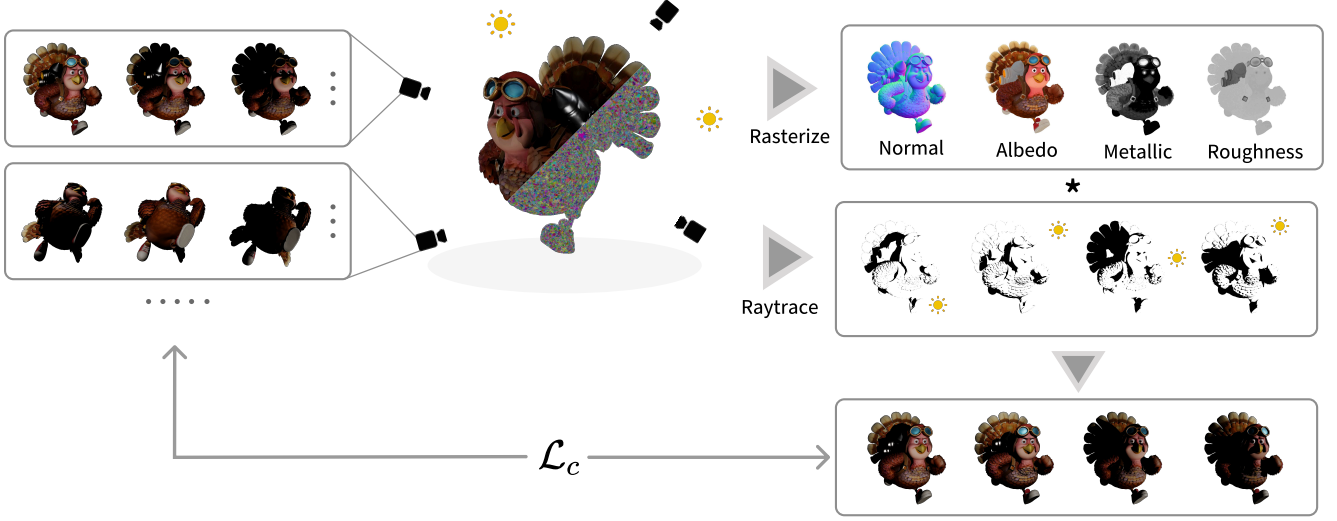


Figure 1. **Stage II pipeline.** After standard 2DGS training, we replace spherical harmonics with per-surfel PBR attributes. For a given viewpoint, we rasterize material and geometry maps into G-buffers and trace shadow rays to each OLAT light source via our Raytracing module. The resulting visibility maps and G-buffers are combined (*) through deferred shading to produce the final rendered images, which are supervised against the corresponding OLAT captures.

2.2. Neural Surface Reconstruction

Accurate geometry is a prerequisite for physically plausible inverse rendering. NeuS [17] introduced an unbiased volume rendering formulation for signed distance functions, enabling high-quality mesh extraction without explicit supervision, though at the cost of rendering efficiency. SuperNormal [18] accelerates neural SDF reconstruction by leveraging multi-view normal maps, producing geometry comparable to 3D scanners and underscoring the value of explicit normal supervision. 2DGS [9] takes a complementary approach, collapsing volumetric Gaussians into planar, surface-aligned disks that enforce geometric consistency while retaining the efficiency of Gaussian splatting. We build upon 2DGS, using its surfel representation as the foundation for joint optimization of geometry and PBR materials.

2.3. Inverse Rendering

Inverse rendering decomposes scene observations into geometry, materials, and illumination. Differentiable rendering has enabled joint optimization from multi-view images, progressing from foundational neural methods such as NeRV [3] and PhySG [21] to nvdiffric [16], which bridges neural representations with traditional graphics pipelines via differentiable marching tetrahedra.

Multi-view photometric stereo (MVPS) provides an additional cue through varying illumination. PS-NeRF [19] pioneered neural MVPS by combining radiance fields with photometric stereo constraints and shadow-aware differentiable rendering. PS-GS [5] improved efficiency by adopt-

ing 2DGS with a learned lighting MLP and ray-traced visibility regularization.

Gaussian splatting has also been extended to general inverse rendering. GaussianShader [10] attaches BRDF properties to individual Gaussians; GS3 [2] decomposes appearance into diffuse, specular, and residual Gaussian fields; and Relightable Neural Gaussians [6] incorporates microfacet BRDFs with precomputed visibility. However, standard 3D Gaussians lack the crisp surfaces necessary for accurate BRDF evaluation, and computing physical shadows via ray-scene intersection negates the efficiency of rasterization.

Our method addresses both limitations through the surface-aligned representation of 2DGS and a fully differentiable BVH-accelerated ray tracing module for shadow modeling. Unlike PS-GS, which relies on a learned lighting network and uncalibrated regularizers, i2DGS exploits known OLAT illumination with explicit visibility computation, enabling physically grounded hard shadow modeling without precomputation or heuristic approximations.

3. Method

Our pipeline takes multi-view OLAT images with known camera and point light positions as input and recovers geometry along with per-surfel PBR materials (albedo a , roughness r , metalness m). We build on 2D Gaussian Splatting (2DGS) [9] and adopt a two-stage strategy: geometry initialization under uniform illumination (Stage I), followed by joint geometry and material optimization under OLAT supervision (Stage II).

3.1. Preliminaries

2D Gaussian Splatting. 2DGS represents a scene as a set of oriented planar disks (surfels). Each primitive is defined by a center $\mathbf{p}$, tangent vectors $\mathbf{t}_u, \mathbf{t}_v$, scales s_u, s_v , opacity o , and color coefficients. The surface normal follows directly as $\mathbf{n} = \mathbf{t}_u \times \mathbf{t}_v$. Pixel colors are obtained by alpha-blending sorted primitives along each ray:

$$C = \sum_{i \in \mathcal{N}} c_i \alpha_i T_i, \quad T_i = \prod_{j < i} (1 - \alpha_j), \quad (1)$$

where $\alpha_i = o_i \cdot G_i(u(\mathbf{x}))$ combines opacity with the Gaussian evaluation at the ray-surfel intersection. We refer to [9] for the full ray-splat intersection derivation.

Physically-Based Shading. We replace the spherical harmonics color model with a deferred shading architecture [8]. Per-pixel material attributes are rasterized into G-buffers and shaded using a Disney-style BRDF [4] comprising a GGX specular lobe and Frostbite’s energy-conserving diffuse term [12]. Under point lighting, the outgoing radiance at a pixel is:

$$L_o = \sum_i \frac{V_i \cdot I_i}{4\pi d_i^2} (f_d + f_s) (\mathbf{n} \cdot \mathbf{l}_i) + A \cdot \mathbf{a} \cdot (1 - m), \quad (2)$$

where V_i is visibility, d_i the light distance, I_i the light intensity, and A a global ambient term.

3.2. Stage I: Geometry Initialization

We train a standard 2DGS model on uniformly lit (full-brightness) images to establish robust initial geometry. Depth and normal maps are obtained by alpha-blending per-primitive values with normalized weights $w_i = T_i \alpha_i / \sum_j T_j \alpha_j$. We optimize using the photometric loss from [9]:

$$\mathcal{L}_c = (1 - \lambda_s) \|\hat{I} - I\|_1 + \lambda_s (1 - \text{SSIM}(\hat{I}, I)), \quad (3)$$

combining ℓ_1 reconstruction with structural similarity. We also apply the depth distortion and normal consistency regularizers of [9]; additional regularization details are provided in the supplementary material.

We inherit the adaptive density control of 3DGS [11] with the homodirectional gradient criterion of [20]. Denoising statistics and optimizer state carry over to Stage II.

3.3. Stage II: Inverse Rendering with OLAT Supervision

3.3.1. Material Modeling

Each surfel is augmented with learnable PBR parameters: albedo $\mathbf{a} \in [0, 1]^3$, roughness $r \in [0, 1]$, and metalness

$m \in [0, 1]$. Per-pixel material maps are obtained via alpha-blending:

$$\{\mathbf{A}, R, M, \mathbf{N}\} = \sum_i w_i \{\mathbf{a}_i, r_i, m_i, \mathbf{n}_i\}. \quad (4)$$

These G-buffers are shaded per-pixel using Eq. (2).

3.3.2. Differentiable Visibility via BVH Ray Tracing

Accurate shadow computation is critical for disentangling materials from lighting. We tessellate each surfel into a triangle fan and build an OptiX BVH over the resulting geometry. For each pixel, the rasterizer outputs intersection points $\{x_k\}$ with blending weights $\{w_k\}$. We trace shadow rays from every intersection toward each light source and accumulate visibility as:

$$V(x, l) = \prod_j (1 - \alpha_j(x, l)), \quad V_{\text{pixel}} = \sum_k w_k \cdot V(x_k, l), \quad (5)$$

where α_j is the opacity of occluders along the shadow ray. To limit overhead, we adaptively cap the intersection count per ray by tracking the 95th percentile across the scene.

3.3.3. Loss Functions

Stage II retains the photometric objective $\mathcal{L}_c$ evaluated against OLAT images rendered via Eq. (2), and adds a *shadow sharpness loss* that encourages binary visibility:

$$\mathcal{L}_s = \frac{1}{|\Omega|} \sum_{p \in \Omega} V_p (1 - V_p). \quad (6)$$

The total Stage II objective is:

$$\mathcal{L}_{\text{II}} = \mathcal{L}_c + \lambda_s \mathcal{L}_s. \quad (7)$$

Loss weights are scheduled with a curriculum that establishes coarse structure before activating detail-level terms. Further regularization details are provided in the supplementary material.

4. Experiments

We evaluate i2DGS along two axes: geometric reconstruction quality and relighting fidelity. All experiments use four synthetic objects from the Trellis 2 objects, each captured under multi-view OLAT conditions with known camera poses and point light positions.

4.1. Geometric Evaluation

We compare reconstructed meshes against ground-truth geometry from three baselines: 2DGS [9] and NeuS [17], both trained on fixed (uniform) lighting, and SuperNormal [18], which leverages OLAT supervision. We report Chamfer Distance (CD), F1 Score at a 0.25% threshold, and

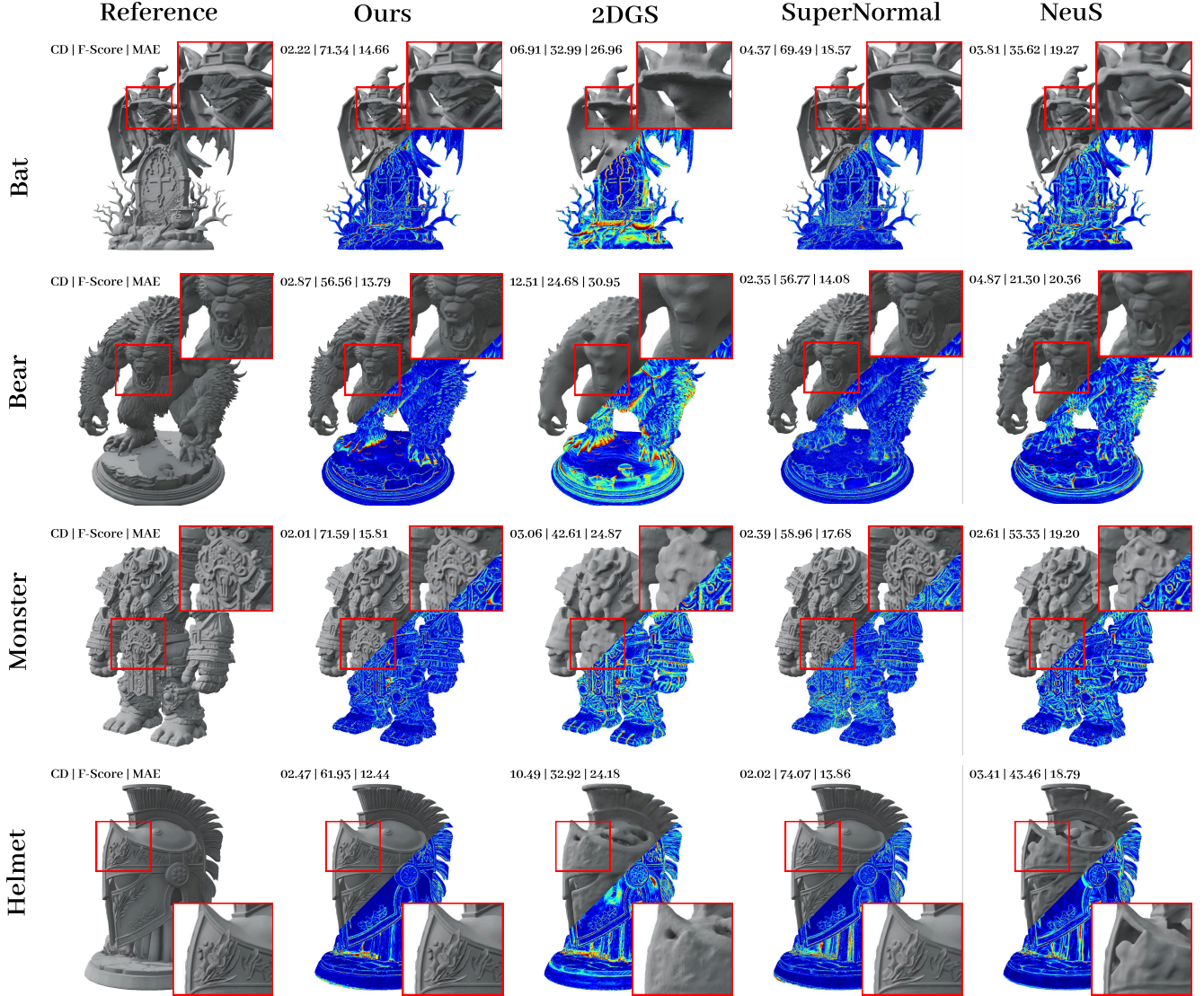


Figure 2. **Geometric reconstruction results.** Qualitative and quantitative comparison on four Trellis 2 objects. i2DGS achieves the lowest Chamfer Distance, highest F1 Score, and lowest MAE across all objects.

Mean Angular Error (MAE) of surface normals. Results are shown in Fig. 2.

i2DGS consistently outperforms all baselines across every metric and object. Compared to 2DGS and NeuS, which optimize geometry purely for view synthesis under fixed illumination, our OLAT-supervised second stage recovers significantly finer surface detail. SuperNormal benefits from the same OLAT cues, yet its implicit SDF representation lacks the explicit surfel alignment that allows i2DGS to maintain sharp geometric features.

4.2. Relighting Evaluation

We assess material reconstruction quality through a relighting protocol: each recovered model is rendered under 409

novel pose–lighting combinations not seen during training, and the resulting images are compared to ground-truth renders using PSNR, SSIM, and LPIPS. We compare against GS3 [2] and Relightable Neural Gaussians (RNG) [6], both of which attach PBR properties to 3D Gaussians but lack explicit surface alignment and physically grounded shadow computation.

i2DGS substantially outperforms both baselines across all three metrics, confirming that the combination of surfel-based geometry, deferred PBR shading, and BVH-accelerated visibility yields more accurate material estimates. The improvement is particularly pronounced under grazing lighting angles, where correct shadow modeling is essential for faithful reproduction of cast shadows and spec-



Figure 3. **Relighting results.** Qualitative and quantitative comparison under 409 novel pose-lighting conditions on the Trellis 2 dataset. i2DGS achieves the best PSNR, SSIM, and LPIPS across all objects.

ular highlights.

5. Conclusion and Future Work

We presented i2DGS, a novel inverse rendering pipeline for high-fidelity 3D geometry and material reconstruction from multi-view OLAT images. By directly coupling 2D

Gaussian Splatting with explicitly modeled shadows via a fully differentiable, BVH-accelerated ray tracing module, our two-stage approach accurately disentangles surface geometry from physically-based material properties. Evaluations confirm that i2DGS outperforms current state-of-the-art methods in both geometric reconstruction and relighting fidelity, producing PBR assets suitable for downstream ap-

plications.

Future Work. While our method adaptively caps ray intersections to limit computational overhead, future research could focus on further optimizing the ray-tracing module for enhanced efficiency. Additionally, extending the i2DGS framework beyond strictly controlled OLAT configurations and known point light positions to handle uncalibrated, complex natural illumination environments represents a promising next step.

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